-Only a small portion of words in a text play an important role in text mining, even after stop-word removal, stemming and lemmatization. When processing text data the main issues is that the dimensionality is too high (N terms >> N docs) and there can be sparneness problem.

-Dimension reduction is used to make large problems computationally problem efficient (saving storage and network resources), it can reduce the amount of training data while keeping the desired level of performance.

-Process of data analytics: phase 1: learning phase: pre-processing with (normalization, dimension reduction and image processing) -> learning (supervised, unsupervised learning) -> error analysis. Phase 2: prediction: the model generates prediction based on new data.

-Dimensionality reduction is divided into feature selection (selecting only some variables from variable set) and feature extraction (creating new variables). Feature extraction creates a new set of features based on the original variables. Example Z1 = V1 + 0.2*V2

-Filter vs Wrapper approach: Filter method selects features based on pre-defined criteria, while wrapper evaluates a subset with a learning algorithm and repeats it until performance is achieved. Wrapper methods can offer higher accuracy but may take longer than filter method.

-Document frequency DF: calculates the total amount of documents in a witch a word w is presented. Words that appear too often may provide unnecessary information.

-Accuracy Acc: expected accuracy of simple classifier built from the single feature

Acc(w) = N(pos,w) – N(Neg,w), helps improvement of text mining models.

-Accuracy ratio $AccR(w) = \left|\frac{N(Pos,w)}{N(Pos)} - \frac{N(Neg,w)}{N(Neg)}\right|$ evaluates how consistently a word appears in both positive and negative documents.

-Probability ratio matrix: $PR(w) = \frac{N(Pos,w)}{N(Pos)} / \frac{N(Neg,w)}{N(Neg)}$ the probability of the word given divided by the probability of the word given the negative class. Evaluates how strongly a word is associated with a specific class.

-Odds ratio (OddR) $OddR(w) = \frac{N(Pos,w)}{N(Neg,w)} \times \frac{N(Neg,\overline{w})}{N(Pos,\overline{w})}$ Reflect the odds of the word occurring in the positive class normalized by that of the negative class. Used to rank relevance.

-Odds ratio Numerator (OddN) : $OddN(w) = N(Pos,w) \times N(Neg,\overline{w})$ is used to quantitively evaluation how frequently a word appears in each class.

-F1-Measure : $F1(w) = \frac{2 \times N(Pos,w)}{N(Pos)+N(w)}$ evaluates how well a word is detected in positive class and how accurate that detection is.

-Information Gain: IG measures the decrease in entropy when feature is given vs absent, when entropy is 0 with a word w present, the information gain is at its maximum.

$$IG(w) = Entropy(absent\ w) - Entropy(given\ w)$$

-Chi-squared statistics (X^2) : Measures divergence from the distribution expected if one assumes the feature occurrence is independent of the class label

$$\chi^2(w) = \frac{N \times [P(Pos,w) \times P(Neg,\overline{w}) - P(Neg,w) \times P(Pos,\overline{w})]^2}{P(w) \times P(\overline{w}) \times P(Pos) \times P(Neg)}$$

-Bi-Normal séparation : Measures the degree of separation assuming that the occurrence of a feature in a document is a random process following a normal distribution$BNS(w) = \left|F^{-1}\left(\frac{N(Pos,w)}{N(Pos)}\right) - F^{-1}\left(\frac{N(Neg,w)}{N(Neg)}\right)\right|$